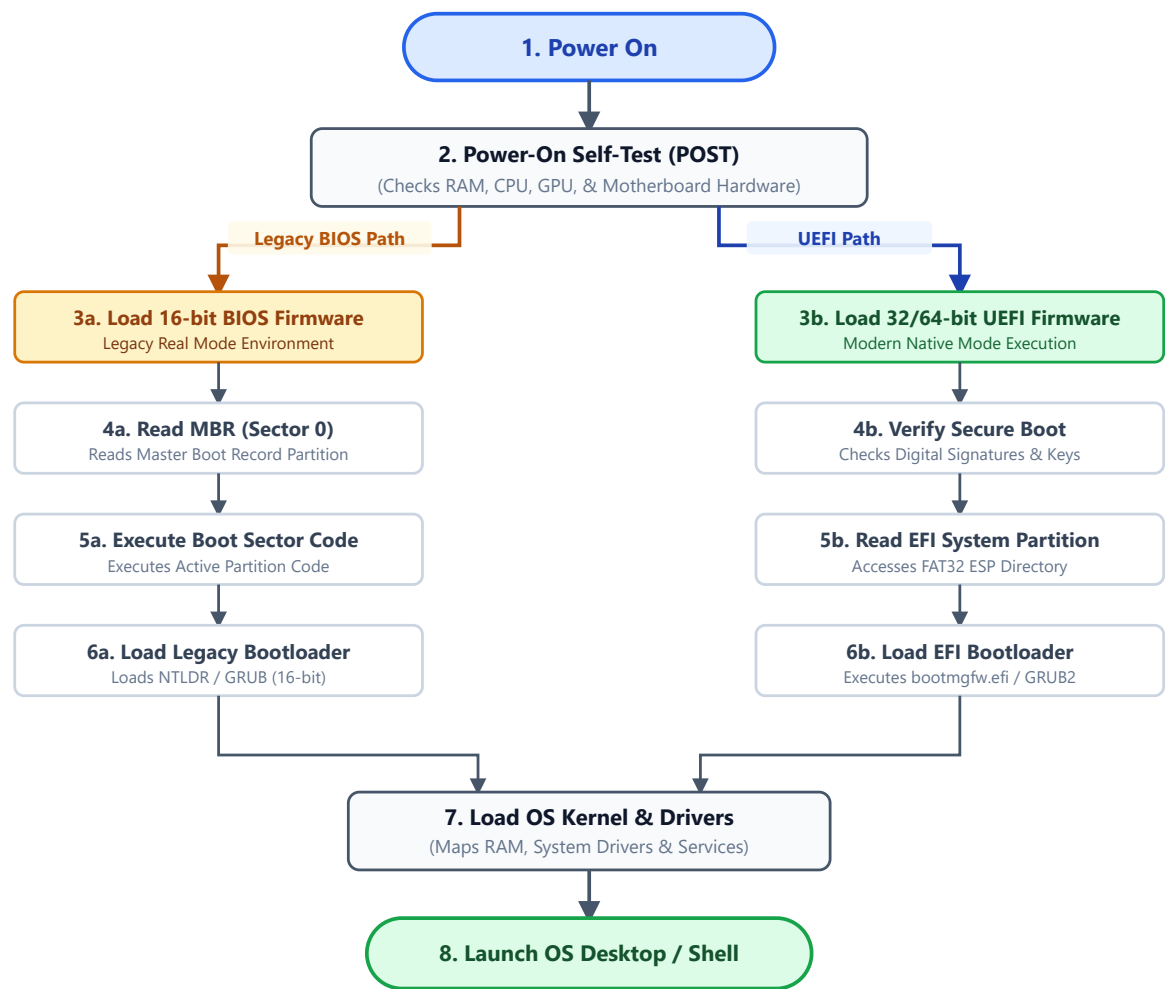


# System Boot Process: BIOS vs. UEFI

System Administration and Maintenance (Flowchart Reference)

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## Boot Sequence Key Highlights

### 1. Firmware Initialization & Hardware POST

- Power-On Self-Test (POST):** Tine-test ng motherboard firmware ang CPU, RAM, display controller, at storage bago magsimula ang boot process.

### 2. Execution Mode & Security Differences

- Legacy BIOS Path:** Gumagana sa 16-bit Real Mode (1MB memory limit). Binabasa ang Master Boot Record (MBR) sa physical Sector 0 ng hard drive.
- UEFI Path:** Gumagana sa 32/64-bit Mode. Binabasa ang FAT32 **EFI System Partition (ESP)** at ginagamitan ng **Secure Boot** para i-verify ang digital signatures ng bootloaders.

### 3. Kernel Handoff & OS Launch

- Kernel Loading:** Iniloload ng bootloader ang OS Kernel sa system RAM, sabay-sabay na inii-initialize ang essential hardware drivers.
- User Shell:** Kapag kumpleto na ang services, lalabas na ang Graphical User Interface (GUI) o Command Line Interface (CLI).